

Consensus Tracking Under Directed Interaction Topologies: Algorithms and Experiments

Wei Ren, *Member, IEEE*

Abstract—Consensus tracking problems with, respectively, bounded control effort and directed switching interaction topologies are considered when a time-varying consensus reference state is available to only a subgroup of a team and the team members have only local interaction. A consensus tracking algorithm explicitly accounting for bounded control effort is proposed and analyzed under a directed fixed interaction topology. Furthermore, convergence analysis for a consensus tracking algorithm is provided when the time-varying consensus reference state is available to a dynamically changing subgroup of the team under directed switching inter-vehicle interaction topologies. Experimental results of a formation control application are demonstrated on a multi-robot platform to validate one of the proposed consensus tracking algorithms.

Index Terms—Consensus, cooperative control, formation control, multi-vehicle systems.

I. INTRODUCTION

COOPERATIVE control of multi-vehicle systems has received significant attention in recent years due to its numerous potential applications in space-based interferometers, combat, surveillance, and reconnaissance systems, hazardous material handling, and distributed reconfigurable sensor networks. Cooperative control often requires that individual vehicles share a consistent view of the objectives and the world. For example, a cooperative rendezvous task requires that each vehicle know the rendezvous point. Consensus algorithms guarantee that vehicles sharing information have a consistent view of information that is critical to the coordination task. The instantaneous value of that information is the information state. To achieve consensus, a vehicle updates the value of its information state based on the information states of its local neighbors. The goal is to design a distributed update law so that the information states of all of the vehicles in the network converge to a common value (see [1], [2], and references therein). Consensus-type techniques have been used to solve formation control problems in [3]–[8] and flocking problems in [9], [10], to name a few.

Current research in consensus algorithms primarily assumes that the consensus equilibrium is a weighted average or a weighted power mean of the initial information states and

therefore constant. This assumption might not be appropriate when each vehicle's information state evolves over time, as occurs in formation maneuvering problems. In addition, many consensus algorithms ensure only that the information states converge to a common value but does not allow specification of a particular value. While this paradigm is useful for applications such as cooperative rendezvous where there is not a single correct value, there are many applications where there is a desired, or reference, information state. In this case, the convergence issues include both convergence to a common value, as well as convergence of the common state to its reference value. Existing consensus-based formation control approaches are often limited to formation stabilization problems, where the vehicles are regulated to constant locations. To guarantee the formation to track a time-varying trajectory, current approaches often rely on the assumption that all vehicles know the time-varying group reference trajectory or velocity. Existing consensus-based flocking algorithms either lack a group reference or require each vehicle to know the group reference. It is hence interesting to study consensus tracking problems when a time-varying reference is available to only a subgroup of a team and the team members have only local interaction.

Consensus with a constant leader state under undirected switching inter-vehicle interaction topologies is addressed in the leader following case of [11]. Consensus with a constant leader state is further considered in [12] and [13] in the context of a directed fixed interaction topology. Dynamic consensus is studied in [14], where an input signal is available to each agent in the team. Estimation algorithms for dynamic average consensus are studied in [15], where a proportional algorithm and a proportional-integral algorithm are analyzed. A consensus problem with a time-varying leader state is solved in [16] under a variable undirected interaction topology, where it is assumed that the leader's acceleration input is available to each follower in the team. In [17], consensus tracking algorithms are proposed and analyzed under a directed fixed interaction topology, where a time-varying *consensus reference state* is available to only a subset of the team members, called the *group leaders*. The consensus tracking algorithms are also applied in [8] to a formation control problem under a directed fixed interaction topology.

While [8] and [17] have addressed the consensus tracking problem with a time-varying consensus reference state, the algorithms in [8] and [17] do not explicitly account for bounded control effort. Furthermore, the convergence analysis in [17] is restricted to the case of a directed fixed interaction topology. In practice, both the group leaders and the directed inter-vehicle interaction topologies may be dynamically switching.

In this brief, we first propose a consensus tracking algorithm to account for bounded control effort and provide convergence analysis in the case of fixed group leaders and a directed fixed

Manuscript received May 04, 2008. Manuscript received in final form February 06, 2009. First published September 01, 2009; current version published December 23, 2009. Recommended by Associate Editor M. Mesbahi. This work was supported by a National Science Foundation CAREER Award (ECCS-0748287).

The author is with the Electrical and Computer Engineering Department, Utah State University, Logan, UT 84322 USA (e-mail: wren@engineering.usu.edu).

Color versions of one or more of the figures in this brief are available online at <http://ieeexplore.ieee.org>.

Digital Object Identifier 10.1109/TCST.2009.2015285

inter-vehicle interaction topology. We then provide convergence analysis for a consensus tracking algorithm in the case of dynamically changing group leaders and directed switching inter-vehicle interaction topologies. Finally, we experimentally implement and validate a consensus tracking algorithm for a formation control problem on our multi-robot platform in the case of dynamically changing group leaders and directed switching inter-vehicle interaction topologies. These results extend the consensus tracking results in [8] and [17]. All of the results in this brief are based on directed interaction topologies. It is worthwhile to mention that an undirected interaction topology is a special case of a directed interaction topology. A preliminary version of this brief was presented at the 2008 American Control Conference [27].

II. GRAPH THEORY NOTIONS

A weighted graph consists of a node set $\mathcal{V} = \{1, \dots, p\}$, an edge set $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$, and a weighted adjacency matrix $A_p = [a_{ij}] \in \mathbb{R}^{p \times p}$. An edge (i, j) in a weighted directed graph denotes that vehicle j can obtain information from vehicle i , but not necessarily vice versa. In contrast, the pairs of nodes in a weighted undirected graph are unordered, where an edge (i, j) denotes that vehicles i and j can obtain information from one another. The weighted adjacency matrix A_p of a weighted directed graph is defined such that a_{ij} is a positive weight if $(j, i) \in \mathcal{E}$, while $a_{ij} = 0$ if $(j, i) \notin \mathcal{E}$. The weighted adjacency matrix A_p of a weighted undirected graph is defined analogously except that $a_{ij} = a_{ji}$, $\forall i \neq j$, since $(j, i) \in \mathcal{E}$ implies $(i, j) \in \mathcal{E}$. If the weights are not relevant, then a_{ij} is set equal to 1 for all $(j, i) \in \mathcal{E}$. In this brief, self edges are not allowed, i.e., $a_{ii} = 0$.

For an edge (i, j) in a directed graph, i is the parent node and j is the child node. A directed path is a sequence of edges in a directed graph of the form $(i_1, i_2), (i_2, i_3), \dots$, where $i_j \in \mathcal{V}$. A directed tree is a directed graph, where every node has exactly one parent except for one node, called the root, which has no parent, and the root has a directed path to every other node. A directed spanning tree of a directed graph is a directed tree that contains all nodes of the directed graph. A directed graph has or contains a directed spanning tree if there exists a directed spanning tree as a subset of the directed graph. That is, there exists at least one node having a directed path to all of the other nodes.

Let the (nonsymmetric) Laplacian matrix $L_p = [\ell_{ij}] \in \mathbb{R}^{p \times p}$ be defined as [18] $\ell_{ii} = \sum_{j=1, j \neq i}^p a_{ij}$ and $\ell_{ij} = -a_{ij}$, $i \neq j$. The matrix L_p satisfies the conditions

$$\ell_{ij} \leq 0, \quad i \neq j, \quad \sum_{j=1}^p \ell_{ij} = 0, \quad i = 1, \dots, p. \quad (1)$$

For an undirected graph, L_p is symmetric positive semi-definite. However, L_p for a directed graph is not necessarily symmetric. In both the undirected and directed cases, 0 is an eigenvalue of L_p with the associated eigenvector $\mathbf{1}_p$, where $\mathbf{1}_p$ is a $p \times 1$ column vector of all ones.

Given a matrix $S = [s_{ij}] \in \mathbb{R}^{p \times p}$, the directed graph of S , denoted by $\Gamma(S)$, is the directed graph on p nodes $i, i \in \{1, 2, \dots, p\}$, such that there is an edge in $\Gamma(S)$ from node j to

node i if and only if $s_{ij} \neq 0$ (cf. [19]). Again, we assume that there is no self edge (i, i) .

III. CONSENSUS TRACKING ALGORITHMS

Suppose that there are n vehicles in a team and the information states of all vehicles satisfy single-integrator kinematics given by

$$\dot{\xi}_i = u_i, \quad i = 1, \dots, n \quad (2)$$

where $\xi_i \in \mathbb{R}^m$ is the information state of the i th vehicle and $u_i \in \mathbb{R}^m$ is the control input.

The objective of this brief is to design distributed control laws for u_i , $i = 1, \dots, n$, such that the information states of all vehicles converge to a time-varying consensus reference state with, respectively, bounded control effort and directed switching interaction topologies when the time-varying consensus reference state is available to only a subgroup of the team and the team members have only local interaction.

Suppose that the consensus reference state, denoted by ξ^r , satisfies

$$\dot{\xi}^r = f(t, \xi^r) \quad (3)$$

where $f(\cdot, \cdot)$ is piecewise continuous in t and locally Lipschitz in ξ^r .

We introduce a virtual vehicle with the states ξ^r and $\dot{\xi}^r$, named vehicle $n + 1$ without loss of generality. Let $A_{n+1} = [a_{ij}] \in \mathbb{R}^{(n+1) \times (n+1)}$ be the weighted adjacency matrix for the $n + 1$ vehicles, denoting information flow among vehicles and whether a vehicle has access to the consensus reference state. In particular, $a_{ij} > 0$, $i, j = 1, \dots, n$, if vehicle i receives information from vehicle j , $a_{i(n+1)} > 0$ if ξ^r and $\dot{\xi}^r$ are available to vehicle i , and $a_{(n+1)j} = 0$, $j = 1, \dots, n + 1$. Note that $a_{ii} = 0$, $\forall i$.

Before moving on, we need the following lemma.

Lemma 3.1: [20] Suppose that $L_p \in \mathbb{R}^{p \times p}$ satisfies the property (1). Then the following conditions are equivalent: 1) L_p has a simple zero eigenvalue with an associated eigenvector $\mathbf{1}_p$ and all of the other eigenvalues have positive real parts; 2) the directed graph of L_p has a directed spanning tree; 3) the rank of L_p is $p - 1$.

A. Bounded Control Inputs

We first consider consensus tracking with bounded control inputs. We propose a consensus tracking algorithm that guarantees that the maximum control effort is independent of the initial information states $\xi_i(0)$ as

$$u_i = \frac{1}{\eta_i} \left[\sum_{j=1}^n a_{ij} \hat{\xi}_j + a_{i(n+1)} \dot{\xi}^r \right] - \frac{1}{\eta_i} \gamma \tanh \left[\sum_{j=1}^n a_{ij} (\xi_i - \xi_j) + a_{i(n+1)} (\xi_i - \xi^r) \right] \quad (4)$$

where $i = 1, \dots, n$, a_{ij} , $i = 1, \dots, n$, $j = 1, \dots, n + 1$, is the (i, j) th entry of the weighted adjacency matrix A_{n+1} , $\eta_i \triangleq \sum_{j=1}^{n+1} a_{ij}$, $\hat{\xi}_j$ is the estimate of ξ_j , γ is a constant positive scalar, and $\tanh(\cdot)$ is defined componentwise. Note that

a_{ij} , $i, j = 1, \dots, n$, specify whether an interaction link from vehicle j to vehicle i exists, and $a_{i(n+1)}$, $i = 1, \dots, n$, specify whether vehicle i has access to ξ^r and $\dot{\xi}^r$.

Note that in (4) each vehicle's control input depends on its local neighbors' information states and their derivative estimates because of the tracking nature of (4). Here $\hat{\xi}_j$ plays an important role in tracking the time-varying consensus reference state in a distributed manner. In practical implementation, the derivatives can be estimated by numerical differentiation of the local neighbors' information states. Next, we will show conditions under which each control input is bounded and independent of the initial information states, and consensus tracking is achieved under a directed fixed interaction topology.

Theorem 3.1: Let $\epsilon_j \triangleq \hat{\xi}_j - \xi_j$ and $\delta \triangleq [\delta_1^T, \dots, \delta_n^T]^T$ with $\delta_i \triangleq \sum_{j=1}^n a_{ij} \epsilon_j$. Suppose that $\|\delta\|$ is uniformly upper bounded by $\sqrt{mn}\gamma\theta$, where γ is defined in (4) and $0 < \theta < 1$, $f(t, \xi^r)$ in (3) is bounded, and the directed graph of A_{n+1} has a directed spanning tree.¹ Using (4) for (2), u_i is bounded, $\|u_i\|_\infty$ is independent of $\xi_i(0)$, and the closed-loop system is input-to-state stable with $\xi_i - \xi^r$ being the state and δ_i being the input. In addition, when $\delta_i(t) \rightarrow 0$, $i = 1, \dots, n$, as $t \rightarrow \infty$, $\xi_i(t) \rightarrow \xi^r(t)$, $i = 1, \dots, n$, as $t \rightarrow \infty$.

Proof: Noting that all entries of the last row of A_{n+1} are zero, it follows that the virtual vehicle $n+1$ with the states ξ^r and $\dot{\xi}^r$ does not have any incoming interaction link. As a result, if the directed graph of A_{n+1} has a directed spanning tree, then each vehicle except the virtual vehicle $n+1$ has at least one incoming interaction link, which implies that $\eta_i \neq 0$, $i = 1, \dots, n$. Thus, (4) is well defined.

Let $e_i \triangleq \sum_{j=1}^n a_{ij}(\xi_i - \xi_j) + a_{i(n+1)}(\xi_i - \xi^r)$. Note that (4) can be written as

$$\left(\sum_{j=1}^{n+1} a_{ij} \right) u_i = \sum_{j=1}^n a_{ij} \dot{\xi}_j + \sum_{j=1}^n a_{ij} \epsilon_j + a_{i(n+1)} \dot{\xi}^r - \gamma \tanh(e_i),$$

which implies

$$\dot{e}_i = -\gamma \tanh(e_i) + \delta_i \quad (5)$$

by noting that $\dot{\xi}_i = u_i$. It thus follows that (5) can be written in matrix form as

$$(L_{n \times (n+1)} \otimes I_m) \underline{u} = -\gamma \tanh(e) + \delta \quad (6)$$

where $\underline{u} \triangleq [u_1^T, \dots, u_n^T, \dot{\xi}^r]^T$, $e \triangleq [e_1^T, \dots, e_n^T]^T$, $\otimes$ denotes the Kronecker product (see [21] for its properties), I_m denotes the $m \times m$ identity matrix, and $L_{n \times (n+1)}$ is an $n \times (n+1)$ matrix with $\ell_{ii} = \sum_{j=1, j \neq i}^{n+1} a_{ij}$ and $\ell_{ij} = -a_{ij}$, $i, j = 1, \dots, n$, and $\ell_{i(n+1)} = -a_{i(n+1)}$, $i = 1, \dots, n$.

Let $L_{(n+1) \times (n+1)} = \begin{bmatrix} L_{n \times (n+1)} \\ 0_{1 \times (n+1)} \end{bmatrix}$, where $0_{1 \times (n+1)}$ denotes the $1 \times (n+1)$ zero matrix. Note that according to the definition of $L_{(n+1) \times (n+1)}$, the directed graph of $L_{(n+1) \times (n+1)}$ has a directed spanning tree if and only if the directed graph of A_{n+1} has a directed spanning tree. Also note that $L_{(n+1) \times (n+1)}$ satisfies the property (1) with $p = n+1$. From the arguments (ii)

¹Equivalently, the virtual vehicle $n+1$ has a directed path to all vehicles 1 to n .

and (iii) of Lemma 3.1, it follows that $\text{rank}(L_{(n+1) \times (n+1)}) = n$ if and only if the directed graph of A_{n+1} has a directed spanning tree. This in turn implies that $\text{rank}(L_{n \times (n+1)}) = n$ if and only if the directed graph of A_{n+1} has a directed spanning tree because all entries in the last row of $L_{(n+1) \times (n+1)}$ are zero. Rewrite $L_{n \times (n+1)} = [W|b]$, where $W = [w_{ij}] \in \mathbb{R}^{n \times n}$ is given as $w_{ij} = -a_{ij}$, $i \neq j$, and $w_{ii} = \sum_{j=1, j \neq i}^{n+1} a_{ij}$, and $b \in \mathbb{R}^{n \times 1}$ is given as $b = [-a_{1(n+1)}, \dots, -a_{n(n+1)}]^T$. Noting that $L_{n \times (n+1)}$ has $n+1$ columns and each of its row sum is zero, it follows that the last column of $L_{n \times (n+1)}$ is dependent on its first n columns, where $b = -W\mathbf{1}_n$. As a result, it follows that $\text{Rank}(W) = \text{Rank}([W|b]) = n$ if and only if the directed graph of A_{n+1} has a directed spanning tree.

Note that (6) can be written as

$$(W \otimes I_m)u + (b \otimes I_m)\dot{\xi}^r = -\gamma \tanh(e) + \delta \quad (7)$$

where $u = [u_1^T, \dots, u_n^T]^T$. Because W has full rank and therefore is invertible if and only if the directed graph of A_{n+1} has a directed spanning tree, the boundedness of u_i comes from the fact that b , $\dot{\xi}^r$, $\tanh(\cdot)$, W^{-1} , and δ are all bounded. In particular, we have that $u = \mathbf{1}_n \otimes \dot{\xi}^r - \gamma(W^{-1} \otimes I_m) \tanh(e) + (W^{-1} \otimes I_m)\delta$ because $W^{-1}b = -\mathbf{1}_n$. It is straightforward to see that $\|u_i\|_\infty$ is independent of $\xi_i(0)$ because $\|\tanh(\cdot)\|_\infty \leq 1$.

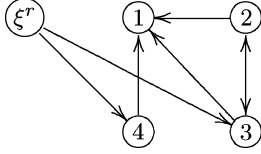
Consider a positive-definite function $V = \mathbf{1}_{mn}^T \log(\cosh(e))$, where $\log(\cosh(\cdot))$ is defined componentwise. It follows from (5) that the derivative of V is given by $\dot{V} = \dot{e}^T \tanh(e) = -\gamma \|\tanh(e)\|^2 + \delta^T \tanh(e) \leq -\gamma(1 - \theta) \|\tanh(e)\|^2 - \gamma \theta \|\tanh(e)\|^2 + \|\delta\| \|\tanh(e)\|$. Noting that $|\tanh(x)| \leq 1$, $\forall x \in \mathbb{R}$, and $\|\delta\| < \sqrt{mn}\gamma\theta$, it follows that $\dot{V} \leq -\gamma(1 - \theta) \|\tanh(e)\|^2$ if $\|\tanh(e)\| > (\|\delta\|/\gamma\theta)$. Therefore, it follows from Theorem 4.19 in [22] that the closed-loop system of (2) using (4) is input-to-state stable with e being the state and δ being the input.

Note that $e = (W \otimes I_m)\xi + (b \otimes I_m)\xi^r$, where $\xi \triangleq [\xi_1^T, \dots, \xi_n^T]^T$. Also note from the previous argument that $W^{-1}b = -\mathbf{1}_n$ if and only if the directed graph of A_{n+1} has a directed spanning tree. It follows that $\xi - \mathbf{1}_n \otimes \xi^r = (W^{-1} \otimes I_m)e$. Similarly, the closed-loop system of (2) using (4) is input-to-state stable with $\xi - \mathbf{1}_n \otimes \xi^r$ being the state and δ being the input, which in turn implies the first statement of the theorem. When $\delta_i(t) \rightarrow 0$, $i = 1, \dots, n$, as $t \rightarrow \infty$, it follows from the property of input-to-state stability that $\xi_i(t) \rightarrow \xi^r(t)$, $i = 1, \dots, n$, as $t \rightarrow \infty$. ■

Remark 3.2: From the proof of Theorem 3.1, we can see that a conservative bound for $\|u\|_\infty$ is given by $\|\dot{\xi}^r\|_\infty + \gamma \|W^{-1} \otimes I_m\|_\infty + \|W^{-1} \otimes I_m\|_\infty \|\delta\|_\infty$.

Example 3.3: To illustrate, we compare (4) with a consensus tracking algorithm in [17] given as

$$u_i = \frac{1}{\eta_i} \left[\sum_{j=1}^n a_{ij} \hat{\xi}_j + a_{i(n+1)} \dot{\xi}^r \right] - \frac{1}{\eta_i} \gamma \left[\sum_{j=1}^n a_{ij} (\xi_i - \xi_j) + a_{i(n+1)} (\xi_i - \xi^r) \right] \quad (8)$$

Fig. 1. Directed graph of A_{n+1} used in (4) and (8).

where a_{ij} , η_i , and γ are defined as in (4). Note that (4) explicitly accounts for bounded control inputs while (8) does not. In particular, using (4), the maximum $\|u_i\|_\infty$ depends on A_{n+1} , ξ^r , and $\hat{\xi}_j$ but is independent of $\xi_i(0)$ (i.e., the initial conditions of the information states). Given real actuator constraints, we can choose A_{n+1} in (4) to ensure that the constraints are satisfied. In contrast, using (8), the maximum $\|u_i\|_\infty$ depends not only on A_{n+1} , ξ^r , and $\hat{\xi}_j$ but on $\xi_i(0)$. Using (8), it is possible to choose smaller values of the nonzero weights a_{ij} so that the control effort is below a certain bound. However, when $\xi_i(0)$ changes, the same nonzero weights a_{ij} cannot guarantee the same bound. In contrast, (4) can guarantee the same bound for arbitrary $\xi_i(0)$. Therefore, (4) is useful when consensus tracking errors need to be bounded between prespecified bounds for arbitrary initial information states.

Let $f(t, \xi^r) = \sin(t) \sin(2\xi^r)$ in (3) with $\xi^r(0) = 1$. Let $a_{ij} = 1$, $i, j = 1, \dots, n$, if vehicle i receives information from vehicle j and $a_{i(n+1)} = 1$ if ξ^r and $\dot{\xi}^r$ are available to vehicle i . Also let $\xi(0) = [-4.5, 0, 2.5, 5.5]^T$. Fig. 1 shows the directed graph of A_{n+1} used in (4) and (8). Figs. 2 and 3 show, respectively, the consensus tracking results with (4) and (8), where the top subplot shows ξ_i and ξ^r and the bottom subplot shows u_i . For initial information states $\xi_i(0)$ with large difference, corresponding large control effort is needed using (8). However, using (4), the maximum control effort is independent of $\xi_i(0)$. Of course, the tradeoff is that consensus tracking with (4) may converge more slowly than with (8). In particular, according to Remark 3.2, we can show that using (4) in this example, $|u_i| \leq 4.45$ for arbitrary $\xi_i(0)$ given that $|\epsilon_i| \leq 0.05$. Fig. 2 validates the theoretical bound. Of course, the theoretical bound is fairly conservative.

B. Directed Switching Interaction Topologies

We then consider a consensus tracking algorithm given as

$$u_i = \frac{1}{\eta_i(t)} \sum_{j=1}^n a_{ij}(t) \left[\hat{\xi}_j - \gamma(\xi_i - \xi_j) \right] + \frac{1}{\eta_i(t)} a_{i(n+1)}(t) \left[\dot{\xi}^r - \gamma(\xi_i - \xi^r) \right] \quad (9)$$

where $i = 1, \dots, n$, $a_{ij}(t)$, $i = 1, \dots, n$, $j = 1, \dots, n+1$, is the (i, j) th entry of the weighted adjacency matrix $A_{n+1}(t)$ at time t , γ is a constant positive scalar, $\eta_i(t) \triangleq \sum_{j=1}^{n+1} a_{ij}(t)$, and $\hat{\xi}_j$ is the estimate of ξ_j . Note that (9) generalizes the algorithm proposed in [17] to allow for time-varying or dynamically switching weights $a_{ij}(t)$, $i = 1, \dots, n$, $j = 1, \dots, n+1$.

Similar to (4), each vehicle's control input in (9) depends on its local neighbors' information states and their derivative estimates because of the tracking nature of (9). Next, we will

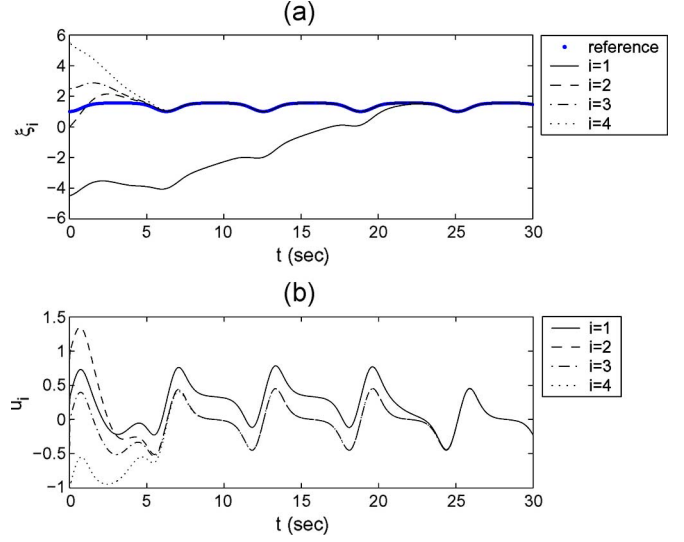


Fig. 2. Consensus tracking with bounded control inputs given by (4). Subplot (a) shows the information states and the consensus reference state while subplot (b) shows the control input.

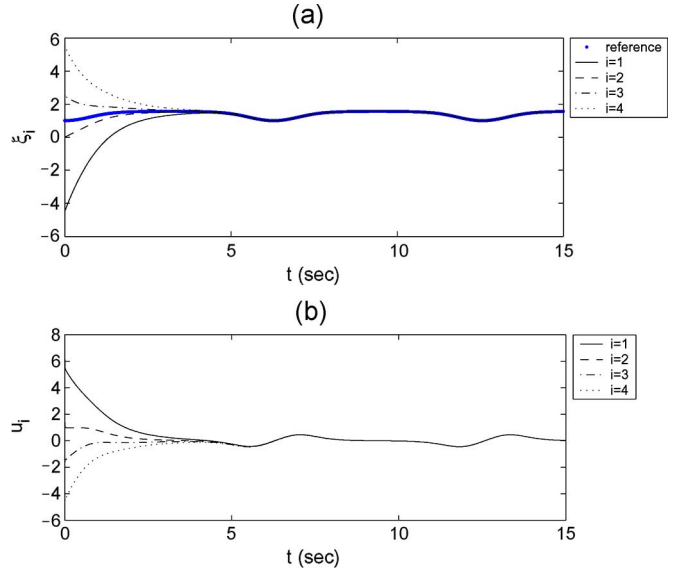


Fig. 3. Consensus tracking without explicitly accounting for bounded control inputs with (8). Subplot (a) shows the information states and the consensus reference state while subplot (b) shows the control input.

show conditions under which consensus tracking is achieved in the case of dynamically changing group leaders under directed switching inter-vehicle interaction topologies.

Theorem 3.4: Let $\epsilon_j \triangleq \hat{\xi}_j - \xi_j$ and $\delta \triangleq [\delta_1^T, \dots, \delta_n^T]^T$ with $\delta_i \triangleq \sum_{j=1}^n a_{ij} \epsilon_j$. Let t_0 be the initial time. Also let $t_1, t_2, \dots$ be the switching times for the directed graph of $A_{n+1}(t) = [a_{ij}(t)] \in \mathbb{R}^{(n+1) \times (n+1)}$ defined in (9). Suppose that $A_{n+1}(t)$ is piecewise continuous and each nonzero (and therefore positive) entry of $A_{n+1}(t)$ is chosen from a compact set $[\underline{a}, \bar{a}]$, where $\underline{a}$ and $\bar{a}$ are positive constants. Also suppose that the directed switching graph of $A_{n+1}(t)$ has a directed spanning tree across each interval $[t_j, t_{j+1})$, $j = 0, 1, \dots$. Using (9) for (2), the closed-loop system is input-to-state stable with $\xi_i - \xi^r$ being the state and δ_i being the input. In addition, when $\delta_i(t) \rightarrow 0$,

$i = 1, \dots, n$, as $t \rightarrow \infty$, $\xi_i(t) \rightarrow \xi^r(t)$, $i = 1, \dots, n$, as $t \rightarrow \infty$.

Proof: The same argument as that in the first paragraph of the proof of Theorem 3.1 can be used to show that $\eta_i(t) \neq 0$, $i = 1, \dots, n$, across each interval $[t_j, t_{j+1})$, $j = 0, 1, \dots$ if the directed switching graph of $A_{n+1}(t)$ has a directed spanning tree across that interval. Thus (9) is well defined across each interval $[t_j, t_{j+1})$, $j = 0, 1, \dots$.

Note that (9) can be written as

$$\begin{aligned} [W(t) \otimes I_m] \dot{u} + [b(t) \otimes I_m] \dot{\xi}^r \\ = -\gamma([W(t) \otimes I_m] \xi + [b(t) \otimes I_m] \xi^r) + \delta \end{aligned} \quad (10)$$

where u , ξ , δ , $W(t)$, and $b(t)$ are defined as in the proof of Theorem 3.1 except that $W(t)$ and $b(t)$ are switching in (10). Following the proof of Theorem 3.1, we can show that $W(t)$ is invertible across each interval $[t_j, t_{j+1})$, $j = 0, 1, \dots$ if and only if the directed graph of $A_{n+1}(t)$ has a directed spanning tree across that interval.

We next show that $W^{-1}(t)$ is bounded under the assumption of the theorem. Noting that each nonzero entry of $A_{n+1}(t)$ is chosen from a compact set, it follows that each nonzero entry of $W(t)$ is also within a compact set. Two matrices are said to have the same structure if their positive, zero, and negative entries are in the same places. Under the assumption that the directed graph of $A_{n+1}(t)$ is switching but has a directed spanning tree across each interval $[t_j, t_{j+1})$, the number of possible directed graphs of $A_{n+1}(t)$ is finite. It then follows that there are a finite number of possible structures for $A_{n+1}(t)$, which implies that there are a finite number of possible structures for $W(t)$. For each possible structure of $W(t)$, $W^{-1}(t)$ exists across each interval $[t_j, t_{j+1})$, which implies that $\det(W(t)) \neq 0$ across each interval $[t_j, t_{j+1})$ if the directed graph of $A_{n+1}(t)$ has a directed spanning tree across that interval. Thus for each possible structure of $W(t)$, $\det(W(t))$ is within a compact set that does not include zero. Also note that for each possible structure of $W(t)$, all entries of its adjoint are within a compact set. It thus follows that each entry of $W^{-1}(t)$ is within a compact set.

Noting that $b(t) = -W(t)\mathbf{1}_n$ and $W(t)$ is invertible across each interval $[t_j, t_{j+1})$ under the assumption of the theorem, it follows that $-W^{-1}(t)b(t) = \mathbf{1}_n$. Noting that $\dot{\xi} = u$, we rewrite (10) as

$$\dot{\xi} - \mathbf{1}_n \otimes \dot{\xi}^r = -\gamma(\xi - \mathbf{1}_n \otimes \xi^r) + [W^{-1}(t) \otimes I_m] \delta. \quad (11)$$

Note that $\dot{\xi} - \mathbf{1}_n \otimes \dot{\xi}^r = -\gamma(\xi - \mathbf{1}_n \otimes \xi^r)$ is an exponentially stable LTI system with $\xi - \mathbf{1}_n \otimes \xi^r$ being the state. Because $\|W^{-1}(t)\|$ is bounded, it follows from [23, Lemma 12.4] that system (11) is input-to-state stable with $\xi - \mathbf{1}_n \otimes \xi^r$ being the state and δ being the input despite the fact that $W(t)$ is switching. The statements of the theorem then follow directly. ■

Remark 3.5: Because the algorithm (9) guarantees distributed tracking of a time-varying reference available to only a subgroup of a team, more stringent conditions are required for convergence under directed switching interaction topologies than those for the standard continuous-time consensus algorithm of the form $\dot{u}_i = -\sum_{j=1}^n a_{ij}(\xi_i - \xi_j)$ as studied in [11], [20], [24] and the distributed weighed averaging algorithm as studied in [25] and [26], where the final consensus equilibrium

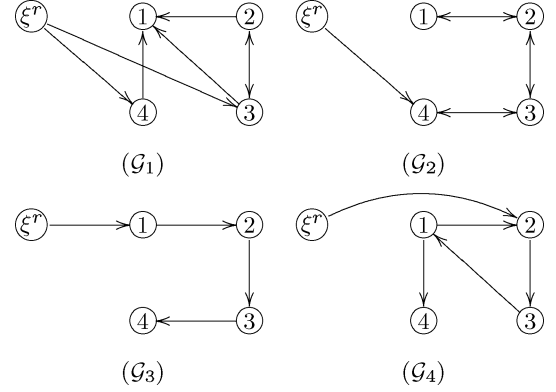


Fig. 4. Directed graphs of $A_{n+1}(t)$ for (9).

is a constant. The above mentioned standard consensus algorithms cannot guarantee tracking of a time-varying reference state that is available to only a subgroup of a team. In addition, while the properties of infinite products of row stochastic matrices are often employed to prove convergence using the above mentioned standard consensus algorithms, these properties are not applicable to the proof for convergence using the consensus tracking algorithm (9).

Remark 3.6: It is worthwhile to mention that a proportional algorithm and a proportional-integral algorithm are proposed in [15] for dynamic average consensus. The objective of the algorithms in [15] is to guarantee that each vehicle can estimate $(1/n) \sum_{i=1}^n u_i$, where u_i is a (possibly time-varying) input signal available to vehicle i . In contrast, the objective of algorithm (9) is to guarantee that each vehicle can track a common reference signal ξ^r when ξ^r is available to only a subset of the vehicles. The results in [15] are not applicable to the consensus tracking problem studied in this brief.

Example 3.7: To illustrate, we assume that the directed graphs of $A_{n+1}(t)$, where $A_{n+1}(t)$ is defined in (9), switch randomly from the set $\bar{\mathcal{G}} = \{\mathcal{G}_1, \dots, \mathcal{G}_4\}$ as shown in Fig. 4 with a switching time around 0.25 s. Fig. 5 shows that each vehicle tracks the consensus reference state as defined in Example 3.3 using (9) under the directed switching interaction topologies shown in Fig. 4.

IV. EXPERIMENTAL VALIDATION ON A MULTI-ROBOT PLATFORM

In this section, we apply one of the consensus tracking algorithms to a formation control problem and experimentally validate the application on a multi-robot platform under directed switching interaction topologies. In particular, four wheeled mobile robots are required to maintain a square formation while their geometric center (GC) is required to follow a circle clockwise.

Let (x_i, y_i) , θ_i , and (v_i, ω_i) denote, respectively, the Cartesian position, orientation, and linear and angular speed of the i th robot. The kinematic equations for the i th robot are

$$\dot{x}_i = v_i \cos(\theta_i), \quad \dot{y}_i = v_i \sin(\theta_i), \quad \dot{\theta}_i = \omega_i. \quad (12)$$

To focus on the main issue, we feedback linearize (12) for a fixed point off the center of the wheel axis denoted as (x_i, y_i) , where

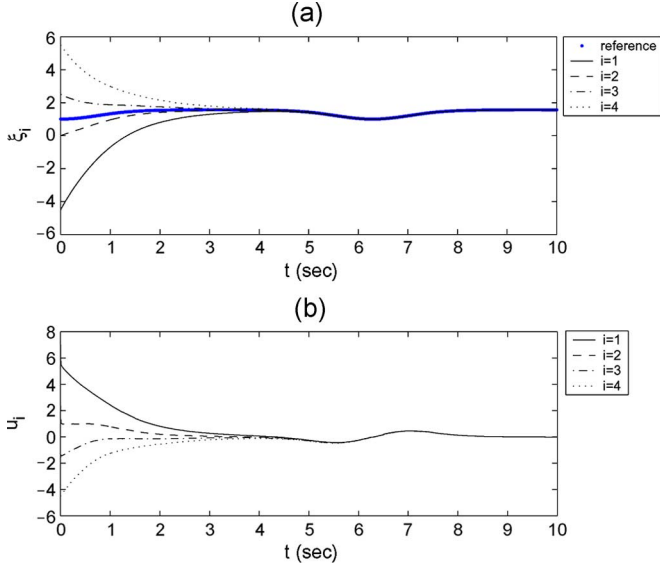


Fig. 5. Consensus tracking with (9). Subplot (a) shows the information states and the consensus reference state while subplot (b) shows the control input.



Fig. 6. Multi-robot experimental platform at Utah State University.

$x_i = r_{xi} + d_i \cos(\theta_i)$ and $y_i = r_{yi} + d_i \sin(\theta_i)$ with $d_i = 0.15$ m. Letting $\begin{bmatrix} v_i \\ \omega_i \end{bmatrix} = \begin{bmatrix} \cos(\theta_i) & \sin(\theta_i) \\ -(1/d_i)\sin(\theta_i) & (1/d_i)\cos(\theta_i) \end{bmatrix} \begin{bmatrix} u_{xi} \\ u_{yi} \end{bmatrix}$, gives $\begin{bmatrix} \dot{x}_i \\ \dot{y}_i \end{bmatrix} = \begin{bmatrix} u_{xi} \\ u_{yi} \end{bmatrix}$, which is a simplified kinematic equation but is sufficient for the purpose of this brief.

In our implementation, a consensus tracking algorithm is applied on the group level, where the robots share their understandings of the geometric center of the team and reach consensus on the time-varying reference trajectory for the geometric center of the team in a distributed manner. Based on the group-level consensus tracking algorithm, a local control law is then applied for robot level control, where the robots are driven to follow their desired positions. As a result, information sharing and consensus occur on the group level rather than on the robot level.

On the group level, let $\xi^r \triangleq [x_c^r, y_c^r, \theta_c^r]$ denote the consensus reference state for the team of robots, where (x_c^r, y_c^r) and θ_c^r represent, respectively, the reference position and orientation for the geometric center of the team. We consider the case where the group leaders that have access to ξ^r and $\dot{\xi}^r$ are dynamically changing and the directed inter-robot interaction topologies are

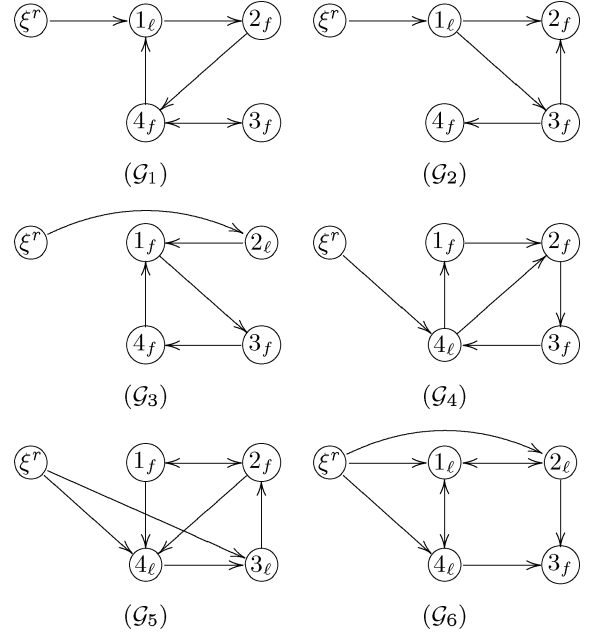


Fig. 7. Directed switching graphs of $A_{n+1}(t)$.

also dynamically changing. Because ξ^r and $\dot{\xi}^r$ are only available to a dynamically changing subgroup of the team, each robot maintains a local variable, $\xi_i \triangleq [x_{ci}, y_{ci}, \theta_{ci}]^T$, which denotes the i th robot's understanding of ξ^r . Here ξ_i is the information state for the i th robot. We apply the consensus tracking algorithm (9) to guarantee that $\xi_i(t)$ tracks $\xi^r(t)$, $i = 1, \dots, n$, for all $\xi_i(0)$. In our experiments, we use Euler method to obtain $\hat{\xi}_j$ on the right side of (9).

On the robot level, each robot determines its desired position (x_i^d, y_i^d) as $\begin{bmatrix} x_i^d \\ y_i^d \end{bmatrix} = \begin{bmatrix} x_{ci} \\ y_{ci} \end{bmatrix} + \begin{bmatrix} \cos(\theta_{ci}) & -\sin(\theta_{ci}) \\ \sin(\theta_{ci}) & \cos(\theta_{ci}) \end{bmatrix} \begin{bmatrix} \hat{x}_{iF} \\ \hat{y}_{iF} \end{bmatrix}$, where $[\hat{x}_{iF}, \hat{y}_{iF}]^T$ represents the desired deviation vector of the i th robot relative to the geometric center of the team. A simple local control law of the form

$$\begin{aligned} u_{xi} &= \dot{x}_i^d - k_{xi}(x_i - x_i^d) \\ u_{yi} &= \dot{y}_i^d - k_{yi}(y_i - y_i^d) \end{aligned} \quad (13)$$

where $k_{xi} > 0$ and $k_{yi} > 0$, is applied to guarantee that $x_i(t)$ tracks $x_i^d(t)$ and $y_i(t)$ tracks $y_i^d(t)$.

With both the group level control law (9) and the vehicle level control law (13), if $\xi_i(t) \rightarrow \xi^r(t)$, $x_i(t) \rightarrow x_i^d(t)$, and $y_i(t) \rightarrow y_i^d(t)$, $i = 1, \dots, n$, as $t \rightarrow \infty$, then the desired formation shape is maintained and the geometric center of the formation follows its reference trajectory.

In our experiment, the consensus reference state $\xi^r = [x_c^r, y_c^r, \theta_c^r]$ satisfies

$$\dot{x}_c^r = v_c^r \cos(\theta_c^r), \quad \dot{y}_c^r = v_c^r \sin(\theta_c^r), \quad \dot{\theta}_c^r = \omega_c^r$$

where $v_c^r = 9\pi/500$ m/s, $\omega_c^r = \pi/50$ rad/s, $(x_c^r(0), y_c^r(0)) = (0, 0)$ m, and $\theta_c^r(0) = 0$ degree. In addition, we let $\hat{x}_{iF} = \ell_i \cos(\phi_i)$ and $\hat{y}_{iF} = \ell_i \sin(\phi_i)$, where $\ell_i = 0.6$ m and $\phi_i = \pi - (\pi/4)i$ rad, $i = 1, \dots, 4$. Thus the lateral length of the square formation is $0.6\sqrt{2}$ m.

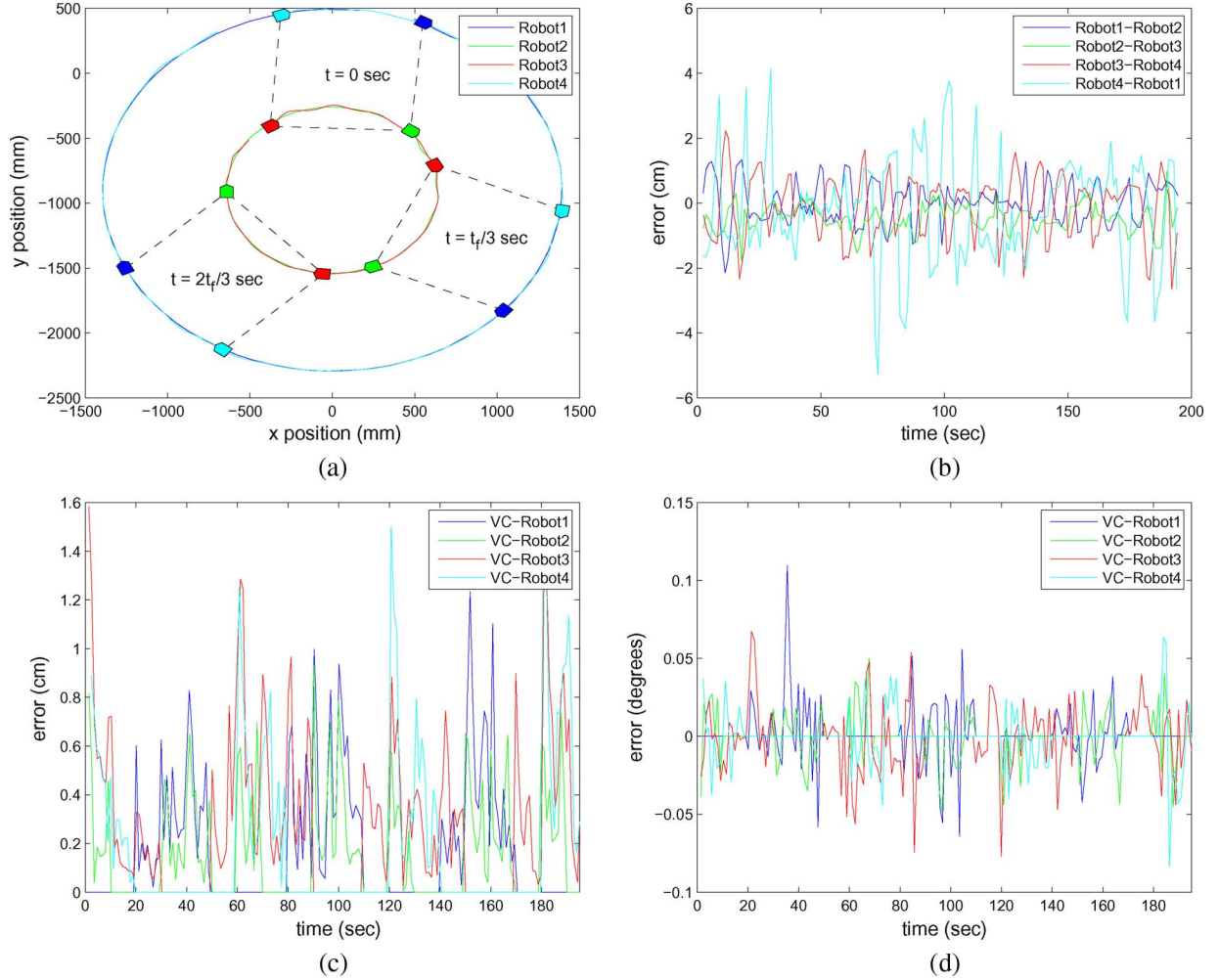


Fig. 8. Consensus tracking experimental results with four mobile robots. (a) Trajectories of the four robots; (b) formation maintenance errors; (c) consensus tracking errors for GC position; (d) consensus tracking errors for GC orientation.

Fig. 6 shows a team of four AmigoBots in the Cooperative Vehicle Networks (COVEN) Laboratory, Utah State University. The robots can communicate with each other through ethernet with TCP/IP protocols. We provide the robots with their initial positions and orientations in a global coordinate frame. The robots then rely on encoder data for their position and orientation measurements.

We assume that the directed graphs of $A_{n+1}(t)$, where $A_{n+1}(t)$ is defined in (9), switch randomly from the set $\bar{\mathcal{G}} = \{\mathcal{G}_1, \dots, \mathcal{G}_6\}$ as shown in Fig. 7 with a switching time around 10 s. In Fig. 7, a subscript ℓ denotes a group leader that has access to ξ^r and ξ^r , a subscript f denotes a follower, and a link from node j to node i denotes that $a_{ij} > 0, i, j = 1, \dots, 4$, in (9). Note that all $\mathcal{G}_i, i = 1, \dots, 6$, have a directed spanning tree with node ξ^r being the root.

Fig. 8(a) shows the trajectories of the four robots at $t \in [0, t_f]$ s and snapshots at $t = 0, t_f/3, 2t_f/3$ s, where t_f is the ending time of the experiment. Note that the four robots maintain a tight formation in our experiment even when both the group leaders and the directed inter-robot interaction topologies are dynamically changing. Fig. 8(b) shows the formation maintenance errors, defined

as $e_{ij} = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2} - \delta_{ij}$, where δ_{ij} is the desired separation distance between robots i and j . That is, e_{ij} represents the difference between the desired and actual separation distance between the robots. Note that the formation maintenance error are between -6 and 5 cm in our experiment. Fig. 8(c) shows the consensus tracking errors for the geometric center position, defined as $\sqrt{(x_c^r - x_{ci})^2 + (y_c^r - y_{ci})^2}$. The consensus tracking errors for the geometric center are below 1.6 cm. Fig. 8(d) shows the consensus tracking errors for the geometric center orientation, defined as $\theta_c^r - \theta_{ci}$. The consensus tracking errors for the geometric center orientation are between -0.1 and 0.15 degrees.

V. CONCLUSION AND FUTURE WORK

We have proposed and analyzed a consensus tracking algorithm accounting for bounded control effort and have shown a convergence condition for a consensus tracking algorithm when the group leaders are dynamically changing and the directed inter-vehicle interaction topologies are switching. We have also applied a consensus tracking algorithm to a formation control problem on our multi-robot experimental

platform. Experimental results have shown the effectiveness of the algorithm even in the presence of dynamically changing group leaders and directed switching inter-robot interaction topologies. The experimental movies of the current brief can be found at <http://www.neng.usu.edu/ece/faculty/wren/videos/Amigobots/multi-leader-experiments.wmv>. While we focus on the first-order kinematics in this brief, the approaches can be extended to more complex dynamics. However, it is expected that more stringent convergence conditions will be required. These will be explored in our future work. In addition, it will be interesting to analyze the convergence condition for the consensus tracking algorithm accounting for bounded control inputs under switching interaction topologies. In this case, the same analysis for the algorithm (9) no longer holds for the algorithm (4). This will also be explored in our future work.

ACKNOWLEDGMENT

The author would like to thank N. Sorensen, L. Ballard, and Y. Cao for their assistance in obtaining the experimental results.

REFERENCES

- [1] W. Ren, R. W. Beard, and E. M. Atkins, "Information consensus in multivehicle cooperative control," *IEEE Control Syst. Mag.*, vol. 27, no. 2, pp. 71–82, Apr. 2007.
- [2] R. Olfati-Saber, J. A. Fax, and R. M. Murray, "Consensus and cooperation in networked multi-agent systems," *Proc. IEEE*, vol. 95, no. 1, pp. 215–233, Jan. 2007.
- [3] J. A. Fax and R. M. Murray, "Information flow and cooperative control of vehicle formations," *IEEE Trans. Autom. Control*, vol. 49, no. 9, pp. 1465–1476, Sep. 2004.
- [4] G. Lafferriere, A. Williams, J. Caughman, and J. J. P. Veerman, "Decentralized control of vehicle formations," *Syst. Control Lett.*, vol. 54, no. 9, pp. 899–910, Sep. 2005.
- [5] M. Porfiri, D. G. Roberson, and D. J. Stilwell, "Tracking and formation control of multiple autonomous agents: A two-level consensus approach," *Automatica*, vol. 43, no. 8, pp. 1318–1328, Aug. 2007.
- [6] D. V. Dimarogonas, S. G. Loizou, K. J. Kyriakopoulos, and M. M. Zavlanos, "A feedback stabilization and collision avoidance scheme for multiple independent non-point agents," *Automatica*, vol. 42, no. 2, pp. 229–243, Feb. 2006.
- [7] D. Lee and M. W. Spong, "Stable flocking of multiple inertial agents on balanced graphs," *IEEE Trans. Autom. Control*, vol. 52, no. 8, pp. 1469–1475, Aug. 2007.
- [8] W. Ren and N. Sorensen, "Distributed coordination architecture for multi-robot formation control," *Robot. Autonomous Syst.*, vol. 56, no. 4, pp. 324–333, Apr. 2008.
- [9] R. Olfati-Saber, "Flocking for multi-agent dynamic systems: Algorithms and theory," *IEEE Trans. Autom. Control*, vol. 51, no. 3, pp. 401–420, Mar. 2006.
- [10] H. G. Tanner, A. Jadbabaie, and G. J. Pappas, "Flocking in fixed and switching networks," *IEEE Trans. Autom. Control*, vol. 52, no. 5, pp. 863–868, May 2007.
- [11] A. Jadbabaie, J. Lin, and A. S. Morse, "Coordination of groups of mobile autonomous agents using nearest neighbor rules," *IEEE Trans. Autom. Control*, vol. 48, no. 6, pp. 988–1001, Jun. 2003.
- [12] K. Moore and D. Lucarelli, "Decentralized adaptive scheduling using consensus variables," *Int. J. Robust Nonlinear Control*, vol. 17, no. 10–11, pp. 921–940, Jul. 2007.
- [13] Z. Jin and R. M. Murray, "Consensus controllability for coordinated multiple vehicle control," presented at the 6th Int. Conf. Cooperative Control Opt., Gainesville, FL, Feb. 2006.
- [14] D. P. Spanos, R. Olfati-Saber, and R. M. Murray, "Dynamic consensus on mobile networks," presented at the IFAC World Congress, Prague, Czech Republic, 2005, Paper Code We-A18-TO/1.
- [15] R. A. Freeman, P. Yang, and K. M. Lynch, "Stability and convergence properties of dynamic average consensus estimators," in *Proc. IEEE Conf. Decision Control*, San Diego, CA, Dec. 2006, pp. 398–403.
- [16] Y. Hong, J. Hu, and L. Gao, "Tracking control for multi-agent consensus with an active leader and variable topology," *Automatica*, vol. 42, no. 7, pp. 1177–1182, Jul. 2006.
- [17] W. Ren, "Multi-vehicle consensus with a time-varying reference state," *Syst. Control Lett.*, vol. 56, no. 7–8, pp. 474–483, Jul. 2007.
- [18] R. Agaev and P. Chebotarev, "On the spectra of nonsymmetric Laplacian matrices," *Linear Algebra Its Applications*, vol. 399, pp. 157–178, 2005.
- [19] R. A. Horn and C. R. Johnson, *Matrix Analysis*. Cambridge, U.K.: Cambridge University Press, 1985.
- [20] W. Ren and R. W. Beard, "Consensus seeking in multiagent systems under dynamically changing interaction topologies," *IEEE Trans. Autom. Control*, vol. 50, no. 5, pp. 655–661, May 2005.
- [21] A. Graham, *Kronecker Products and Matrix Calculus With Applications*. New York: Halsted Press, 1981.
- [22] H. K. Khalil, *Nonlinear Systems*, 3rd ed. Upper Saddle River, NJ: Prentice-Hall, 2002.
- [23] W. J. Rugh, *Linear System Theory*, 2nd ed. Englewood Cliffs, NJ: Prentice-Hall, 1996.
- [24] R. Olfati-Saber and R. M. Murray, "Consensus problems in networks of agents with switching topology and time-delays," *IEEE Trans. Autom. Control*, vol. 49, no. 9, pp. 1520–1533, Sep. 2004.
- [25] L. Moreau, "Stability of multi-agent systems with time-dependent communication links," *IEEE Trans. Autom. Control*, vol. 50, no. 2, pp. 169–182, Feb. 2005.
- [26] V. D. Blondel, J. M. Hendrickx, A. Olshevsky, and J. N. Tsitsiklis, "Convergence in multiagent coordination, consensus, and flocking," in *Proc. IEEE Conf. Decision Control, Eur. Control Conf.*, Seville, Spain, Dec. 2005, pp. 2996–3000.
- [27] W. Ren, "Consensus tracking under directed interaction topologies: Algorithms and experiments," in *Proc. Amer. Control Conf.*, Seattle, WA, Jun. 2008, pp. 742–747.